

# Word Search Visualization

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Asked in: visa , morgan stanley , jpmmc , goldman sachs

## Board & Word

Board:

|   |   |   |   |
|---|---|---|---|
| A | B | C | E |
| S | F | C | S |
| A | D | E | E |

Word: ABCCED

The correct path is **A** → **B** → **C** → **C** → **E** → **D**

## Walkthrough

Now suppose Aadya starts from the first A.

### Step 1

Current board

|   |   |   |   |
|---|---|---|---|
| A | B | C | E |
| S | F | C | S |
| A | D | E | E |

She matches **A**.

Mark it visited.

|   |   |   |   |
|---|---|---|---|
| # | B | C | E |
| S | F | C | S |
| A | D | E | E |

### Step 2

She moves to **B**.

Mark visited.

|   |   |   |   |
|---|---|---|---|
| # | # | C | E |
| S | F | C | S |
| A | D | E | E |

### Step 3

She moves to first **C**.

Mark visited.

|   |   |   |   |
|---|---|---|---|
| # | # | # | E |
| S | F | C | S |
| A | D | E | E |

Now imagine Aadya tries the wrong direction first.

Instead of going down to the second C, she goes **Right → E**

But the next required letter is **C**

This branch fails.

So recursion returns. so hame wapas c ke paas jaane padega and try new branches with new direction , maan lo ek case hota jisme saare direction fail kar jaate to we would have to backtrack even more , so we explore a characters neighbour till all gives false , if all gives false we backtrack ,

## Algorithm

algorithm , try starting from every cell , do a dfs with base case as if  $idx == word.size$  then return true , which means all characters are matched , but if character mismatch branch fails return false, before trying all 4 direction ,, save the board , mark the character seen , try all 4 direction and backtrack ,

## Code Implementation

code is

```
class Solution {
public:

    vector<pair<int,int>> dir={{-1,0},{0,-1},{0,1},{1,0}};
    int n;
    int m;
    bool dfs(vector<vector<char>>& board, string word,int r , int c,int idx){

        if(idx==word.size()){
            return true; //all char match
        }

        if(r<0 || c<0 || r>=n || c>=m) return false;

        if(word[idx]!=board[r][c]){
            return false;
        }
    }
}
```

```

//base SEB rule in dfs backtrack questions
//base cases, seen , explore , backtrack

char temp=board[r][c]; //saving state of board for backtrack
board[r][c]='?';

for(auto &it:dir){ //exploring
    int nr=r+it.first;
    int nc=c+it.second;

    if(dfs(board,word,nr,nc,idx+1)){
        return true;
    }

}

//backtracking

board[r][c]=temp;
return false;

}

bool exist(vector<vector<char>>& board, string word) {
    n=board.size();
    m=board[0].size();

    //har ek cell ko as starting point of dfs choose krenge
    //kyunki maan lo chahiye tha aadya , and start hi d se kiya to dya mil jaayega but

    for(int i=0;i<n;i++){
        for(int j=0;j<m;j++){
            if(dfs(board,word,i,j,0)){ //matching starts from 0th index of word
                return true;
            }
        }
    }

    return false;
}

};

```